

Architecture Decision Record: ADR-003

Implementing AI-to-AI Communication over Zenoh

Abel-Raul Mazilu

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Status: Accepted
Project: DVerse Communication Platform 2.0

1 Context

The DVerse platform is moving toward a multi-agent architecture in which AI agents act as autonomous participants. These agents need a decoupled communication mechanism that does not depend on direct service-to-service calls or the chat application transport layer.

Zenoh has already been selected as the messaging layer for the platform. Its topic-based publish-subscribe model is suitable for asynchronous agent communication and future secured transport requirements.

2 Decision

Sprint 4 will implement AI-agent-to-AI-agent communication over Zenoh. Agents will publish and subscribe to structured Zenoh topics as first-class participants in the system.

This implementation will include the following capabilities:

- Topic conventions for agent discovery, commands, responses, status updates, and errors.
- Structured message schemas for payloads exchanged between agents.
- Publish and subscribe flows that allow agents to communicate without direct coupling.
- Message correlation identifiers for request and response tracking.
- Error handling for failed publications, invalid payloads, and unavailable subscribers.

Zenoh communication must be instrumented with OpenTelemetry spans and Pydantic Logfire logs from the beginning. Each published or consumed message should be observable through both tracing and structured logging.

3 Alternative Decisions

One alternative was to route all agent communication through the chat application. This was rejected because the chat application is a user-facing interface, not the dedicated system messaging layer.

Another alternative was to evaluate a different messaging system during Sprint 4. This was rejected because Zenoh had already been retained from prior architectural decisions due to its performance characteristics, topic-based publish-subscribe model, and suitability for future mTLS/PKI-secured transport.

4 Consequences

4.1 Positive

- Agents can communicate asynchronously without direct coupling.
- The messaging layer remains independent of the chat application interface.
- Topic conventions and message schemas provide a foundation for future agent scaling.
- Zenoh interactions can be traced and logged consistently from the beginning.

4.2 Negative

- Topic naming, schema design, and correlation conventions must be documented clearly.
- Invalid payloads and unavailable subscribers require explicit error-handling behavior.
- Additional integration complexity is introduced during Sprint 4.

5 Scope

Deliverable	Description
Topic conventions	Define Zenoh topics for discovery, commands, responses, status updates, and errors.
Message schemas	Define structured payloads for AI-agent-to-AI-agent communication.
Publish-subscribe flows	Enable agents to exchange messages asynchronously over Zenoh.
Message correlation	Add identifiers for request-response tracking and trace correlation.
Error handling	Handle failed publications, invalid payloads, and unavailable subscribers.

Statement on AI Use

Artificial intelligence was used to support the drafting of parts of this document. The generated text was subsequently reviewed, checked, and edited where necessary by me, the author, Abel-Raul Mazilu.